

Load Flow Analysis of a 1 MW Solar PV Plant Connected to the Grid

Basics of load flow, bus types, and a 4-bus PSS®E case study with simulation results

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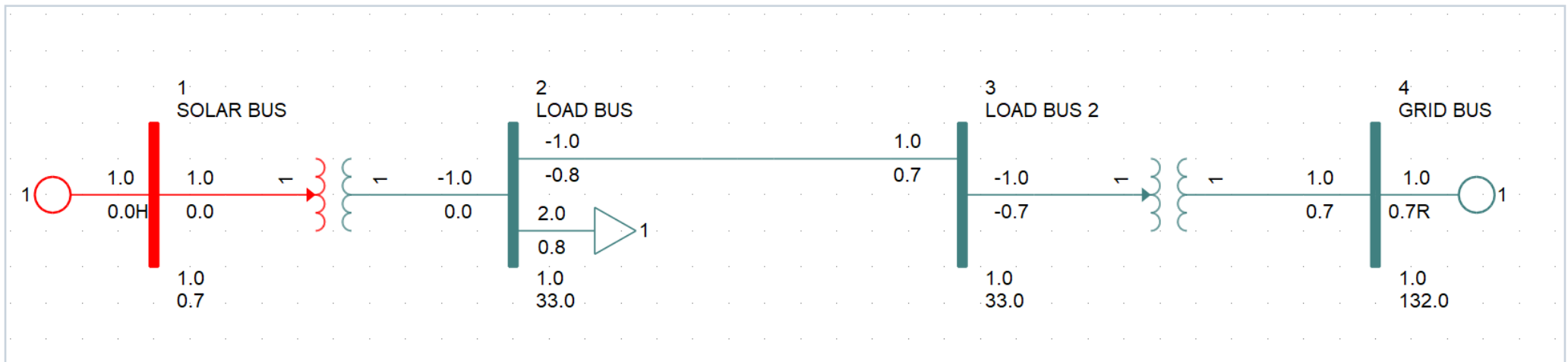
Tool: Siemens PSS®E

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Solved case in PSS®E (solar generator at unity power factor)



Workshop objectives

Workshop Objectives

- Understand the role of PSS®E in power-system studies.
- Understand the basic PSS®E network data model.
- Build a simple 1 MW solar PV plant connected to the grid.
- Create and solve a steady-state power-flow case.
- Interpret bus voltages, generator output, line/transformer loading and losses.
- Develop a workflow that can be extended to larger solar plants.

Slide: Workshop Objectives

Workshop tasks

Perform Load Flow Analysis

Task 1: Develop the simulation case study.

Task 2: Run the load flow assuming a conventional generator.

Task 3: Run the load flow considering a solar generator at unity power factor.

Task 4: Run the load flow considering a solar generator at a different power factor.

Task 5: Adjust the local load to different values (1 + j0.4) MVA, (1.5 + j0.6) MVA, (2 + j0.8) MVA and check the power flow convergence.

Task 6: Check the power flow of a larger power system and interpret the results.

Slide: Perform Load Flow Analysis

1 MW

Solar PV plant

4

Buses in the model

2

Iterations to converge

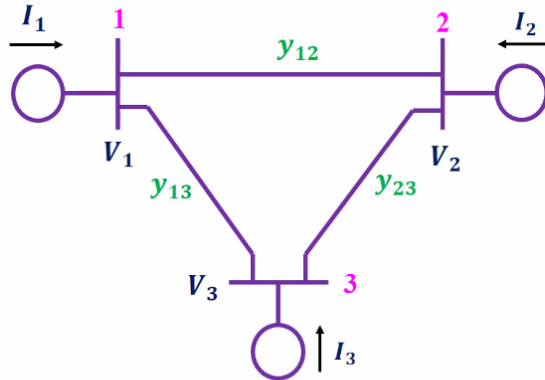
80%

Transformer 1 loading

Scope of this report

- The workshop covered the role of PSS®E in power-system studies, its basic network data model, and building a simple 1 MW solar PV plant connected to the grid.
- This report documents the case I developed (Task 1) and the solved steady-state load flow with the solar generator at unity power factor, including bus voltages, generator output, line/transformer loading and losses.

Theory behind Load Flow Analysis



$$\begin{bmatrix} \underline{I}_1 \\ \underline{I}_2 \\ \underline{I}_3 \end{bmatrix} = \begin{bmatrix} (y_{12} + y_{13}) & -y_{12} & -y_{13} \\ -y_{12} & (y_{12} + y_{23}) & -y_{23} \\ -y_{13} & -y_{23} & (y_{13} + y_{23}) \end{bmatrix} \begin{bmatrix} \underline{V}_1 \\ \underline{V}_2 \\ \underline{V}_3 \end{bmatrix}$$

$$[\underline{I}] = [\underline{Y}][\underline{V}]$$

$$\underline{I}_i = \sum_{k=1}^N Y_{ik} \underline{V}_k \quad (1)$$

- When the system is in steady-state;

$$\sum \text{Generation} = \sum \text{Load} + \text{Loss}$$

- Consider an arbitrary bus in the network;

$$\begin{array}{c} |V_i| \angle \delta_i \\ \downarrow \\ \underline{I}_i \\ \uparrow \\ P_i + jQ_i \end{array}$$

$$\underline{S}_i = \underline{V}_i \underline{I}_i^* \rightarrow \underline{I}_i = \frac{\underline{S}_i^*}{\underline{V}_i^*} = \frac{P_i - jQ_i}{\underline{V}_i^*} \quad (2)$$

$$\frac{P_i - jQ_i}{\underline{V}_i^*} = \sum_{k=1}^N Y_{ik} \underline{V}_k$$

Slide: Theory behind Load Flow Analysis

What a load flow does

- Solves the steady-state operating point of a power system.
- In steady state: **$\Sigma \text{Generation} = \Sigma \text{Load} + \text{Loss}$** .
- Outputs: bus voltages, generator output, line and transformer loading, and losses.
- The bus type (next page) decides which of $|V|$, δ , P and Q are given at each bus.

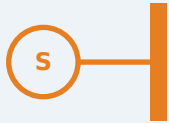
Network equations

- Nodal form: **$[\underline{I}] = [\underline{Y}][\underline{V}]$** , i.e. $I_i = \sum Y_{ik} V_k$ (1)
- Power injected at bus i: $S_i = V_i I_i^*$, so $I_i = (P_i - jQ_i) / V_i^*$ (2)
- Combining (1) and (2): **$(P_i - jQ_i) / V_i^* = \sum Y_{ik} V_k$** , one equation per bus, with Y built from the network data (y_{12} , y_{13} , y_{23} in the slide example).

3 Bus Types in Load Flow Analysis

Swing / slack, PV, PQ

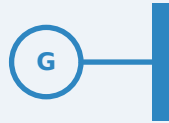
Slack / Swing bus



Establishes the reference angle and balances the system's active-power mismatch.

Reference bus. "Slack" and "swing" are two names for the same bus type.

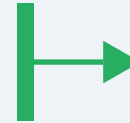
PV bus



Specified active power and voltage magnitude; reactive power is solved within limits.

Voltage-controlled generator bus.

PQ bus



Specified active and reactive power; voltage magnitude and angle are solved.

Load bus, or a generator with fixed P and Q.

What is given and what is solved at each bus type

Bus type	$ V $ voltage magnitude	δ voltage angle	P active power	Q reactive power
Slack / Swing	Specified	Specified	Solved	Solved
PV	Specified	Solved	Specified	Solved
PQ	Solved	Solved	Specified	Specified

Slack: $|V|$ and δ specified, P and Q solved (it takes up the mismatch). PV: P and $|V|$ specified. PQ: P and Q specified.

Bus Types in Load Flow Analysis

- Slack / Swing bus
 - Establishes the reference angle and balances the system's active-power mismatch.
- PV bus
 - Specified active power and voltage magnitude; reactive power is solved within limits.
- PQ bus
 - Specified active and reactive power; voltage magnitude and angle are solved.
- A practical solar-plant model may use a PV-bus representation when voltage control is being studied.
- The selected bus type must reflect the actual control philosophy and operating mode.

Bus	Name	Role in the model
4	GRID BUS 132 kV	Swing bus (confirmed in the solver's SWING BUS SUMMARY)
1	SOLAR BUS 0.69 kV	Generator bus: P = 1.0 MW, Q = 0.0 (unity power factor), so P and Q are solved
2	LOAD BUS 33 kV	Load bus: 2 MW + j0.8 Mvar, V and δ solved
3	LOAD BUS 2 33 kV	Network bus, no load: V and δ solved

Colour code used in this report

Swing / slackreference bus, balances active power

PVP and |V| given (not used in this case)

PQP and Q given

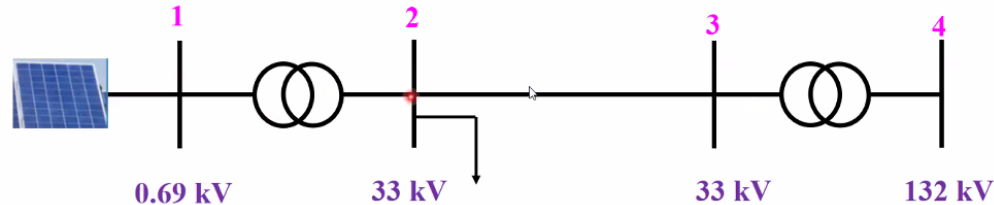
Slide: Bus Types in Load Flow Analysis



Key points from the slide

- A practical solar-plant model may use a PV-bus representation when voltage control is being studied.
- The selected bus type must reflect the actual control philosophy and operating mode of the plant.
- Here the plant runs at unity power factor, so its reactive power is fixed at 0 and the grid swing bus balances the active-power mismatch.

Single Line Diagram of the Test System



Data	
Rated Power	1 MW
Inverter Output	0.69 kV
Transformer 1	0.69/33 kV, 1.25 MVA, 6% on transformer base
Initial Solar Power Factor	1.0
Local Load	2 MW + 0.8 MVar

Data	
Transformer 2	33/ 132 kV, 20 MVA 8% on transformer base
33 kV Feeder	10 km equivalent; R = 0.020 pu, X = 0.040 pu, B = 0.001 pu
Feeder Rating	10 MVA
System Base	100 MVA
Grid Voltage	132 kV (Assume 1.02 pu at the operating point)

Slide: Single Line Diagram of the Test System

System layout

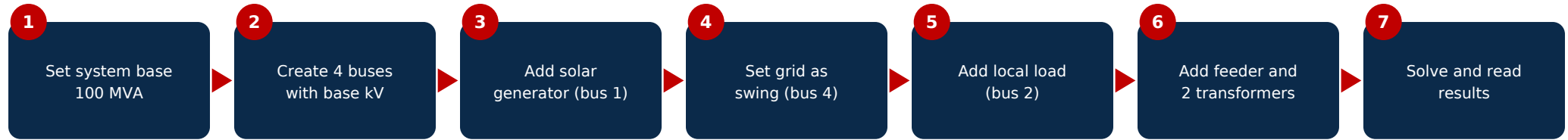
- **Bus 1:** 1 MW solar PV plant, inverter output 0.69 kV.
- **Transformer 1** steps up to 33 kV (Bus 2), where the local load is connected.
- **33 kV feeder** (10 km equivalent) links Bus 2 and Bus 3.
- **Transformer 2** steps up to 132 kV (Bus 4), the grid connection.

Points to note from the data

- Solar plant 1 MW versus Transformer 1 rating of 1.25 MVA.
- Grid voltage 132 kV, assumed 1.02 pu at the operating point.
- Feeder rating 10 MVA; system base 100 MVA.
- Local load 2 MW + j0.8 Mvar is larger than the solar output.

Bus numbering used

- 1 SOLAR BUS (0.69 kV)
- 2 LOAD BUS (33 kV)
- 3 LOAD BUS 2 (33 kV)
- 4 GRID BUS (132 kV)



Model data entered

Bus / plant data	Value
Bus 1 SOLAR BUS	0.69 kV
Bus 2 LOAD BUS	33 kV
Bus 3 LOAD BUS 2	33 kV
Bus 4 GRID BUS	132 kV (swing)
Grid voltage	1.02 pu at operating point
Solar plant	1 MW, initial PF = 1.0
Local load (Bus 2)	2 MW + j0.8 Mvar

Branch	Data
Transformer 1	0.69/33 kV, 1.25 MVA, 6% on transformer base
33 kV feeder	10 km eq.: R = 0.020, X = 0.040, B = 0.001 pu
Feeder rating	10 MVA
Transformer 2	33/132 kV, 20 MVA, 8% on transformer base
System base	100 MVA

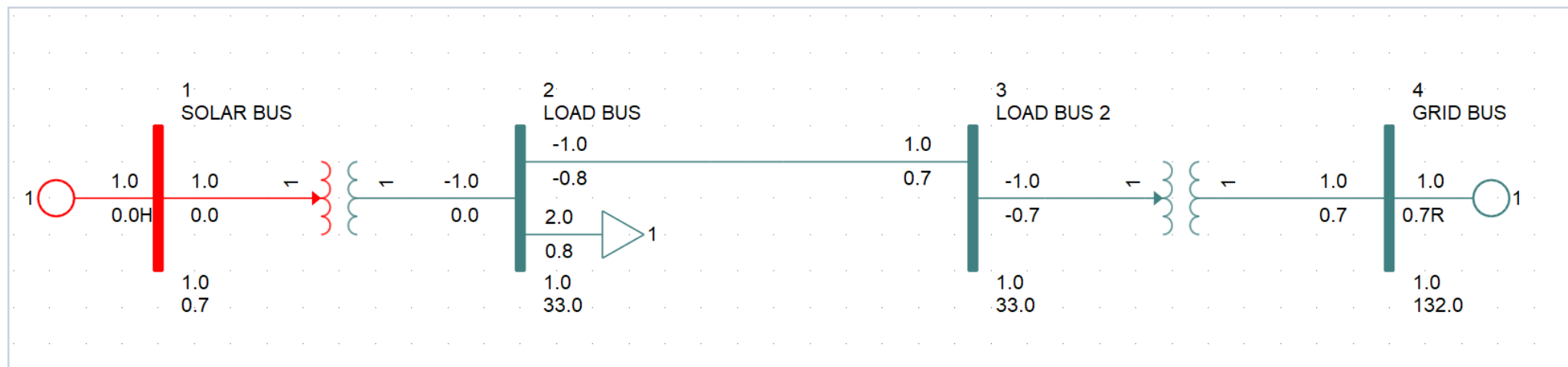
Workshop tasks (Tasks 1 and 3 are documented in this report)

- Task 1: Develop the simulation case study
- Task 2: Run the load flow with a conventional generator
- Task 3: Run the load flow with a solar generator at unity power factor
- Task 4: Run the load flow with a solar generator at a different power factor
- Task 5: Adjust the local load and check convergence
- Task 6: Check the power flow of a larger power system

Source: workshop slides. Feeder R, X, B are in pu on the 100 MVA system base.

7 Results: Solved Bus and Flow Diagram

Simulation output (screenshot)



PSS®E single-line diagram after solving: P (MW) and Q (Mvar) at each end, bus voltage (pu) and base kV

Solar plant (Bus 1)

1.0 MW, 0.0 Mvar

Active power only, consistent with unity power factor.

Local load (Bus 2)

2.0 MW, 0.8 Mvar

Supplied by the solar plant and the grid.

Grid (Bus 4)

1.0 MW, 0.7 Mvar

Swing bus supplies the balance of P and the Q.

Voltages and branch flows read from the diagram

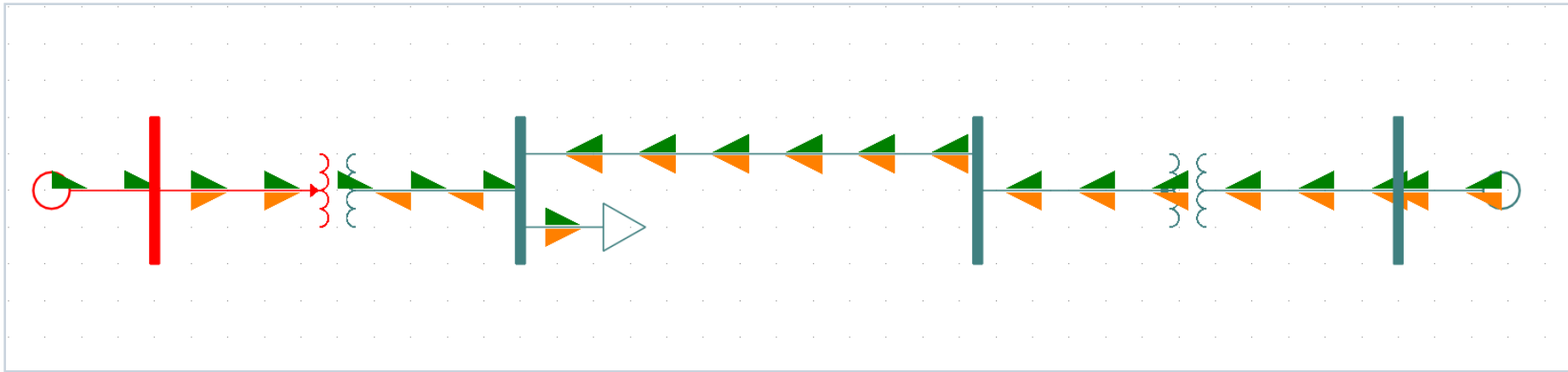
Bus	Name	Base kV	V (pu)
1	SOLAR BUS	0.69	1.0
2	LOAD BUS	33.0	1.0
3	LOAD BUS 2	33.0	1.0
4	GRID BUS	132.0	1.0

Branch	Bus: P / Q (MW / Mvar)	Bus: P / Q (MW / Mvar)
Transformer 1	Bus 1: 1.0 / 0.0	Bus 2: -1.0 / 0.0
33 kV feeder	Bus 3: 1.0 / 0.7	Bus 2: -1.0 / -0.8
Transformer 2	Bus 4: 1.0 / 0.7	Bus 3: -1.0 / -0.7

Voltages are shown to one decimal on the diagram. Positive flow = out of the bus into the branch; negative = into the bus.

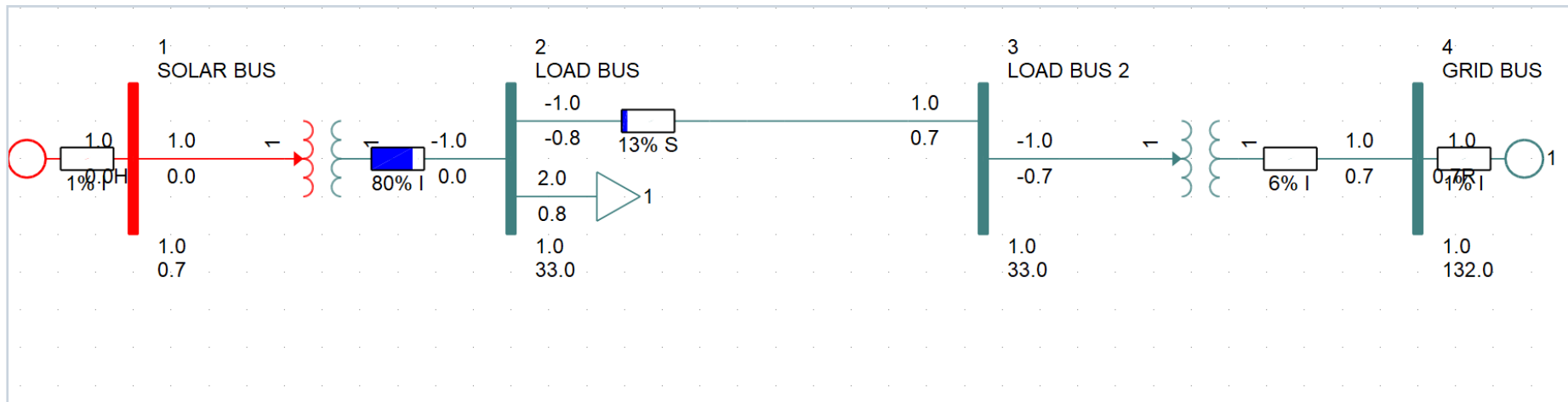
8 Results: Power Flow Direction and Loading

Simulation output (screenshots)



Power flow arrows (green: active power, orange: reactive power)

- Power flows from the grid (Bus 4) through Transformer 2 and the feeder towards Bus 2, and from the solar plant into Bus 2.
- At Bus 2: 1.0 MW (solar) + 1.0 MW (feeder) = 2.0 MW load.
- Reactive flow at the feeder's Bus 2 end (0.8) is 0.1 Mvar above the Bus 3 end (0.7), consistent with the feeder shunt susceptance $B = 0.001$ pu (0.1 Mvar on the 100 MVA base).



Loading (% of rating): solar generator 1%, Transformer 1 80%, feeder 13%, Transformer 2 6%, grid 1%

- Transformer 1 is the most heavily loaded element at 80% (1 MW on a 1.25 MVA rating).
- Feeder loading is 13% of its 10 MVA rating and Transformer 2 loading is 6%; no element is overloaded.

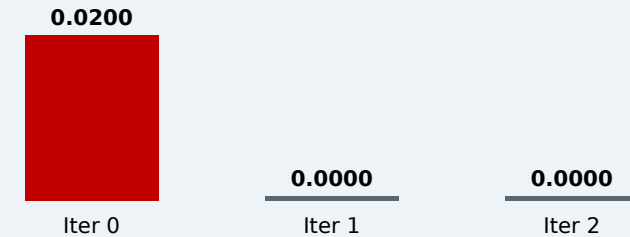
9 Solver Convergence and Power Balance

Iteration log (mismatches in pu, bus in brackets)

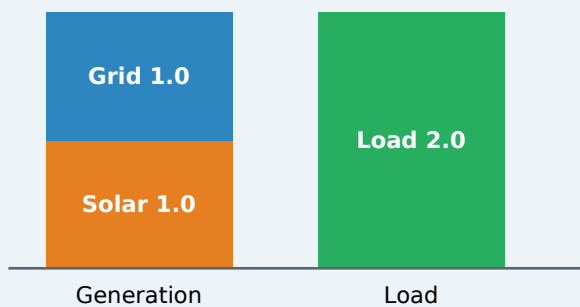
Iter	ΔP	ΔQ	ΔV	ΔAngle
0	0.0200 (2)	0.0075 (2)	0.00106 (1)	0.00105 (2)
1	0.0000 (2)	0.0000 (2)	0.00000 (1)	0.00000 (2)
2	0.0000 (2)	0.0000 (2)	-	-

Reached tolerance in 2 iterations. Largest mismatch 0.00 MW, 0.00 Mvar, 0.00 MVA at bus 2 (LOAD BUS, 33 kV); system total absolute mismatch 0.00 MVA.

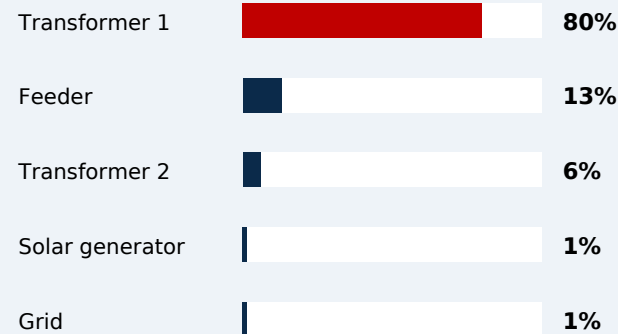
ΔP mismatch per iteration (pu)



Active power balance (MW)



Element loading (% of rating)



Swing bus summary and reading

Bus **4 GRID BUS**
Base kV **132.00**
PGEN **1.0 MW**
QGEN **0.7 Mvar**
P / Q limits **±9999.0**

- Largest initial mismatch: 0.0200 pu (2 MW) at bus 2, equal to the local load.
- Solar 1.0 MW + grid 1.0 MW = load 2.0 MW.

Role of PSS®E

Understood how PSS®E is used in power-system studies.

Network data model

Understood how buses, generators, loads, transformers and branches are entered.

Load flow theory

Nodal equations, steady-state balance, and how bus types define the unknowns.

Modelling a solar plant

Built a 1 MW PV plant connected to the grid through two transformers and a 33 kV feeder.

Solving the case

Solved a steady-state power flow that converged in 2 iterations with zero mismatch.

Interpreting results

Read bus voltages, generator output, transformer/feeder loading and power-flow directions.

Bus types recap

Swing / slack

Bus 4 (GRID BUS): reference angle, balances active power

PV

P and $|V|$ specified; not needed in the unity power factor case

PQ

Buses 1, 2, 3: P and Q specified or load, V and δ solved

Conclusion

- The 1 MW solar plant, feeding a 2 MW + j0.8 Mvar load, was simulated with the grid as swing bus; the grid supplied the remaining 1.0 MW and 0.7 Mvar.
- Transformer 1 was the most loaded element (80%); all other elements were lightly loaded.
- The same workflow (define base, build buses, add generation and load, add branches, solve, interpret) can be extended to larger solar plants, as targeted by the workshop.